

PLM fine-tuning versus standard MLP

Five saved test splits | RMSE comparison | 4 October 2026

Target: $\ln(\text{half-life} + 0.1)$. Lower RMSE is better.

Standard MLP: one-hot peptide + HLA pseudosequence (860 features), hidden layers 256 and 128, dropout 0.2.

Frozen PLM: ESM-2 final-layer features (55,040 inputs) with an MLP.

Fine-tuned PLM: starts from the frozen-ESM MLP, unfreezes ESM-2 last block and trains with regression plus pairwise ranking loss.

Mean test RMSE across five splits:

	All rows	Measured zeros removed
Standard MLP	1.1663	1.1320
Frozen ESM + MLP	1.1718	Not evaluated
Fine-tuned ESM + MLP	1.1626	1.1011

Fine-tuning lowers mean RMSE by 0.31% on all rows and by 2.73% when measured-zero test rows are excluded. Models are not retrained for the positive-only evaluation. The same row mask applies to both.

RMSE is calculated per split, then averaged. Error bars show split SD; overlapping test sets mean this is not a confidence interval.

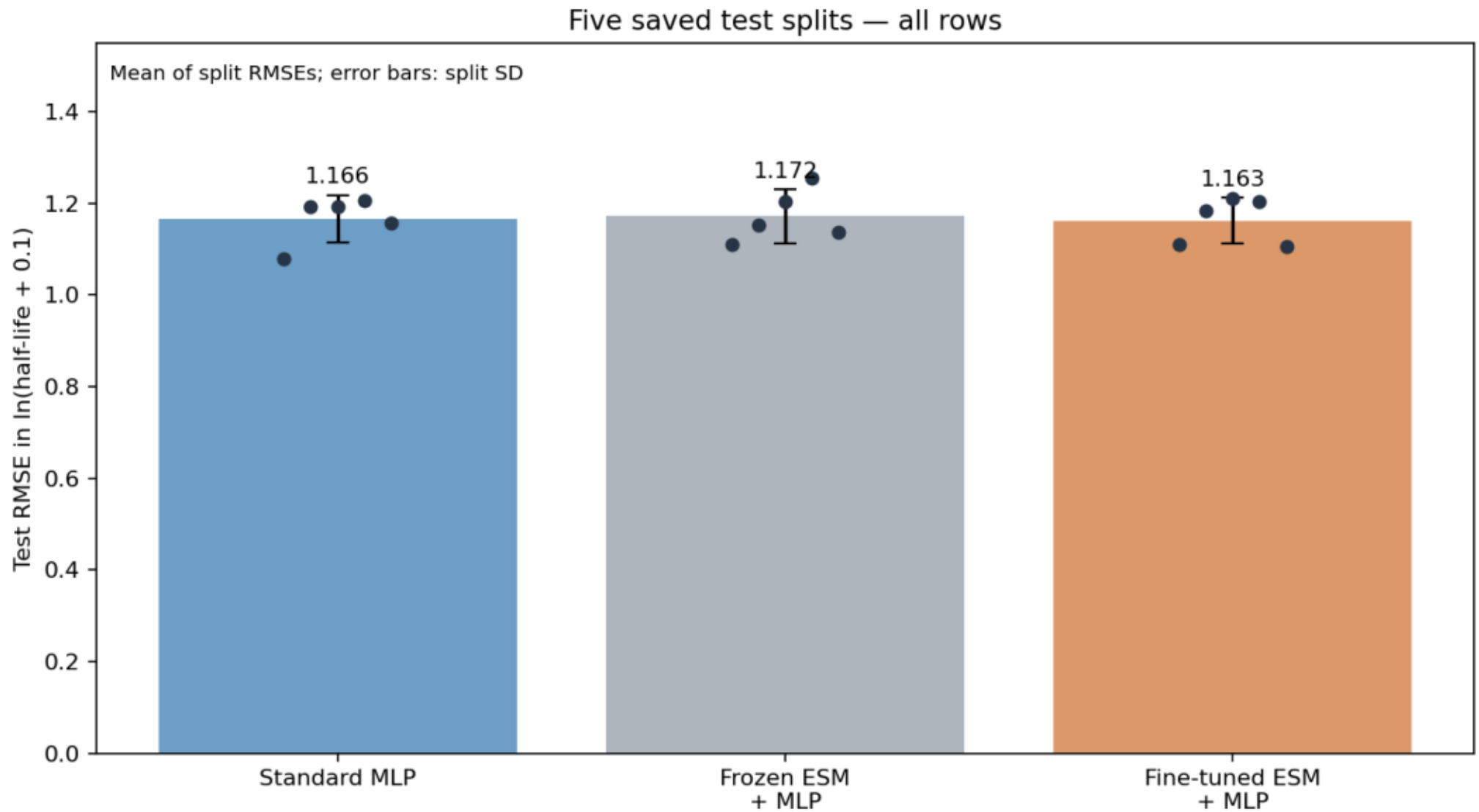
Scatter plots select each model's lowest test RMSE for illustration.

Fine-tuning selected epochs 0, 6, 4, 7, 10 on splits 0 through 4;

split 0 retains its starting checkpoint. No classifier gate is applied.

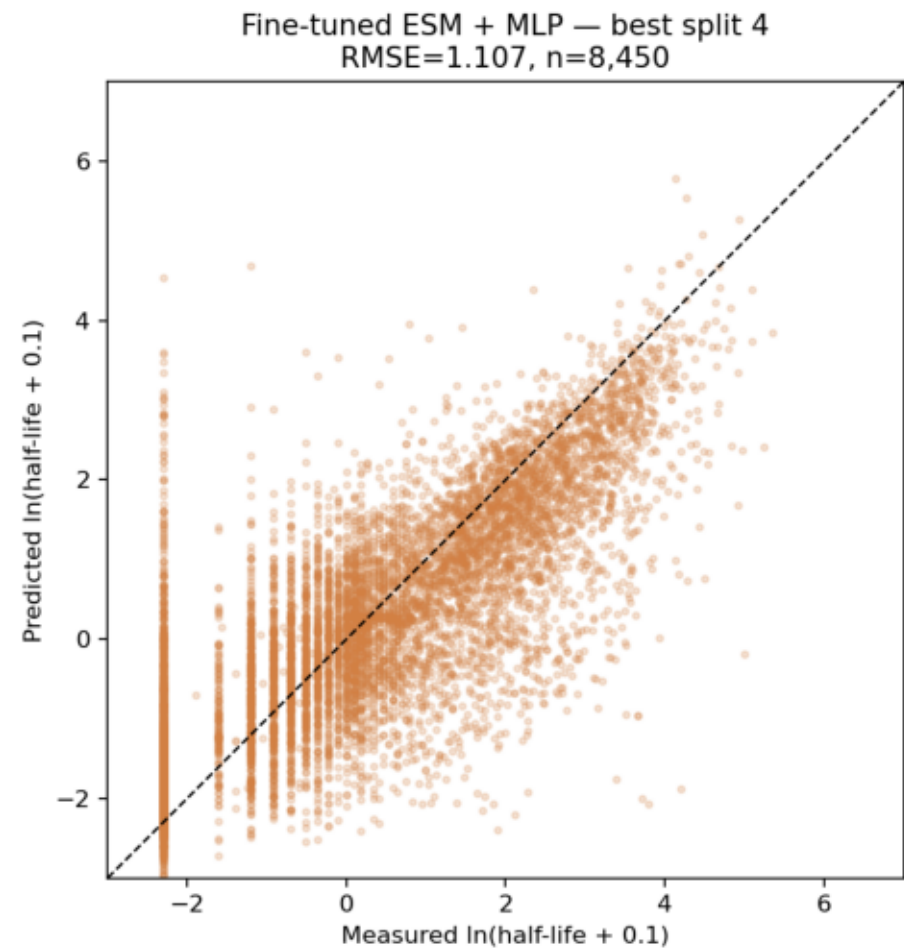
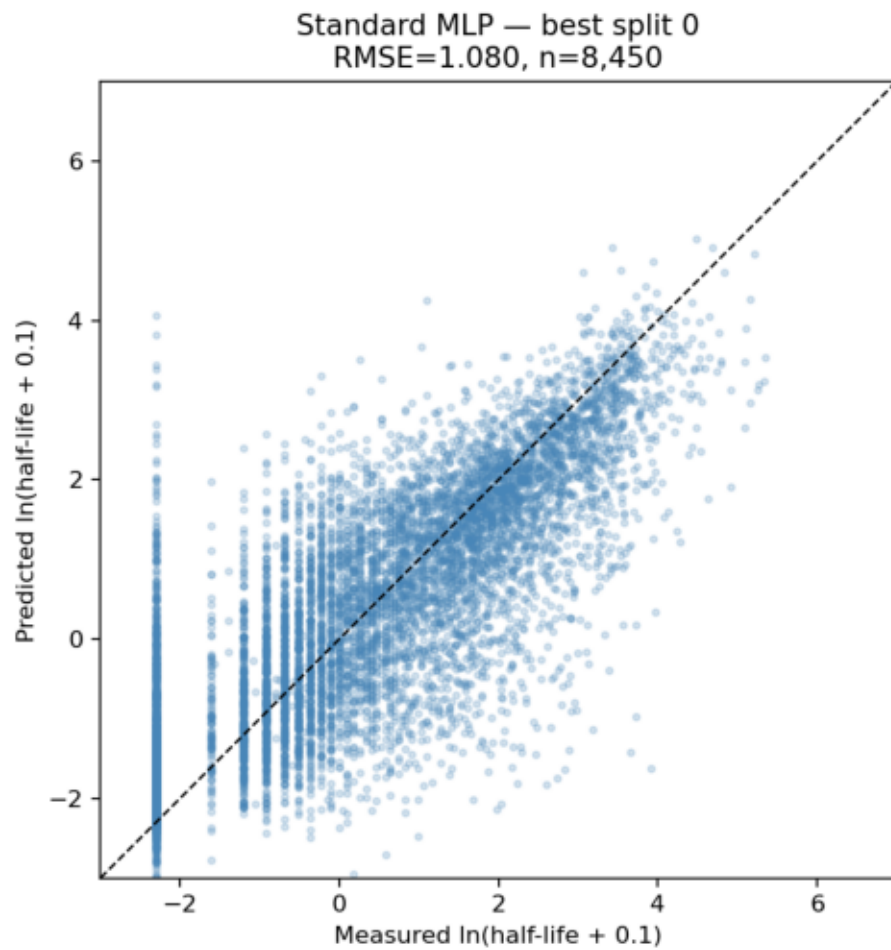
The $k=0.1$ and $k=0.2$ classifier gates increased MSE on every split.

Mean RMSE: all measured half-lives



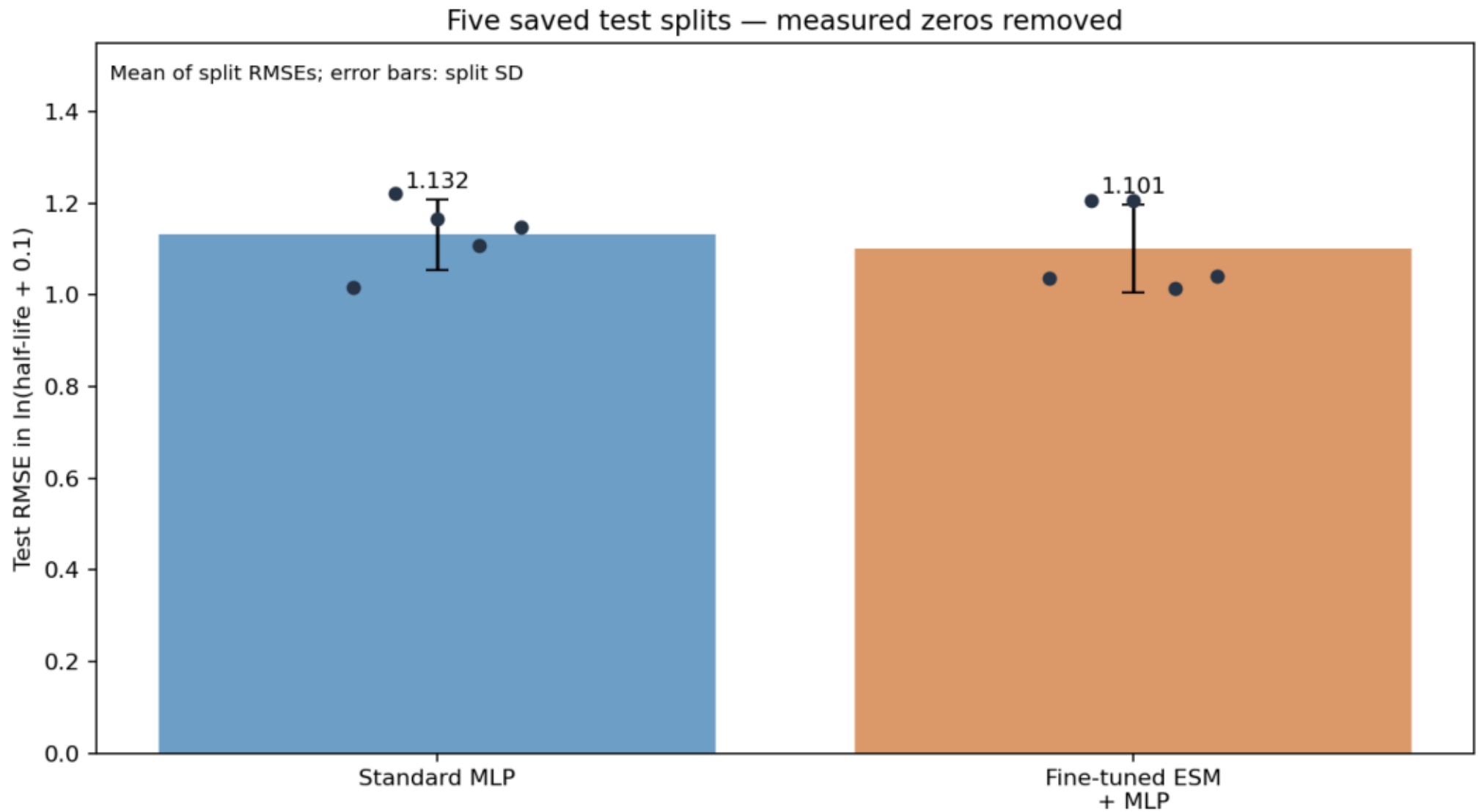
Target: $\ln(\text{half-life} + 0.1)$. Same saved split i for each model. Lower RMSE is better.

Best test scatter plots: all rows



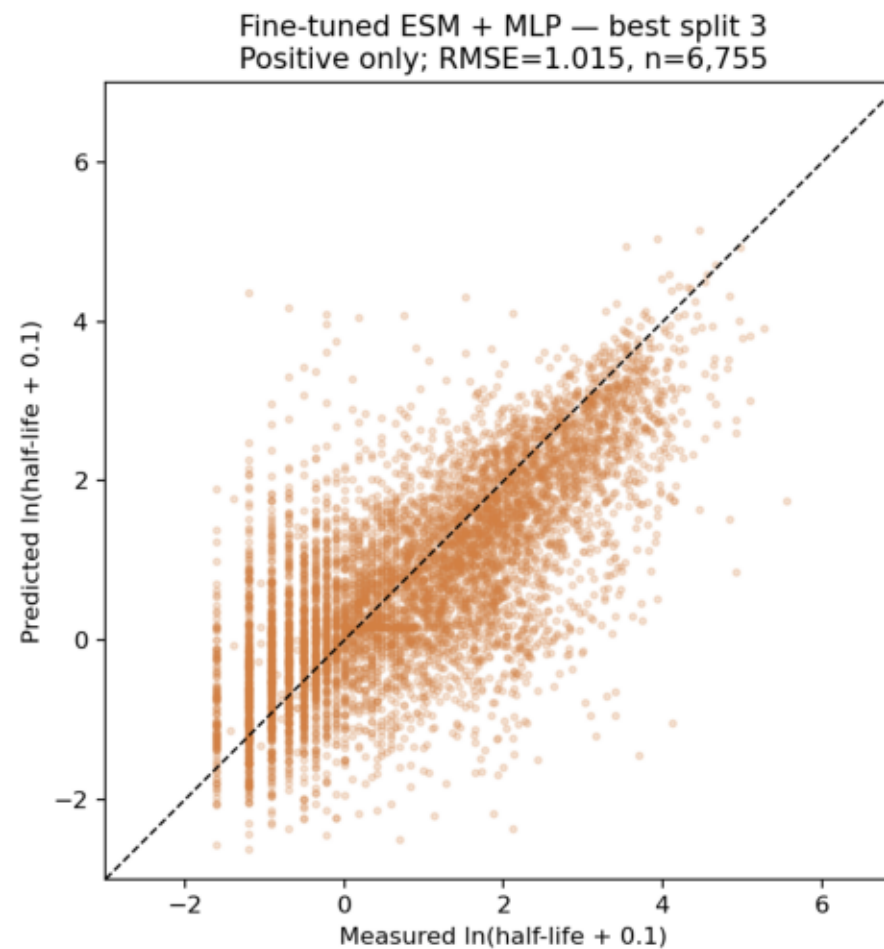
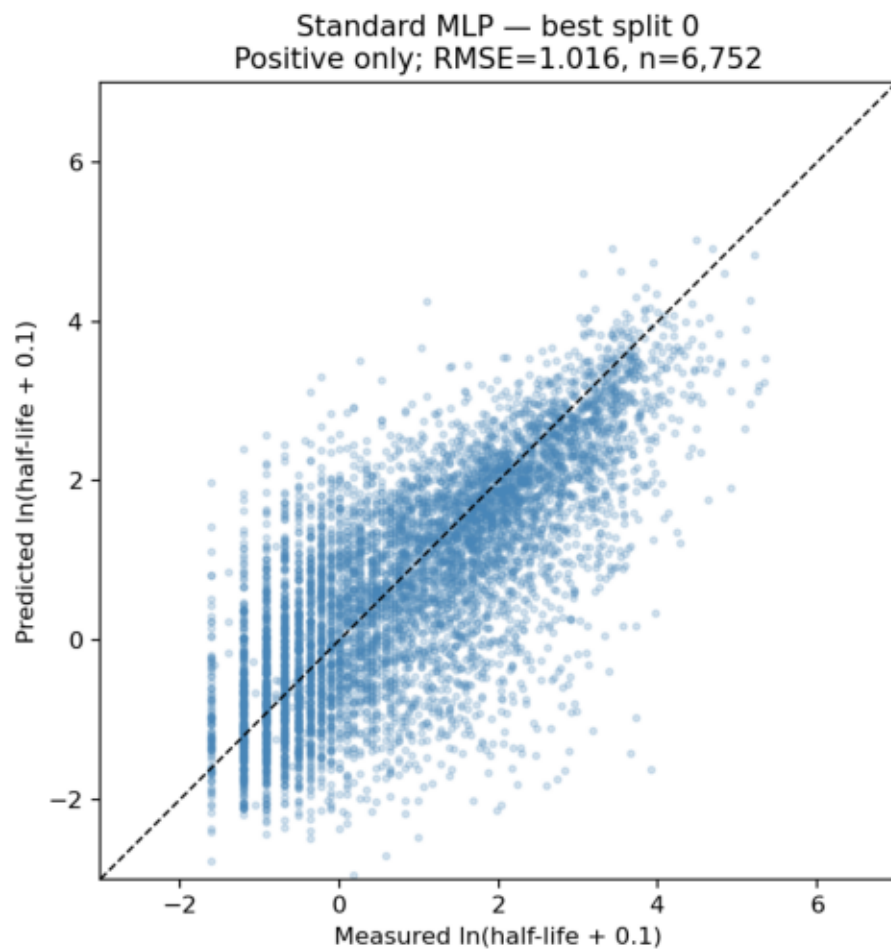
Target: $\ln(\text{half-life} + 0.1)$. Same saved split i for each model. Lower RMSE is better.

Mean RMSE: measured zeros removed



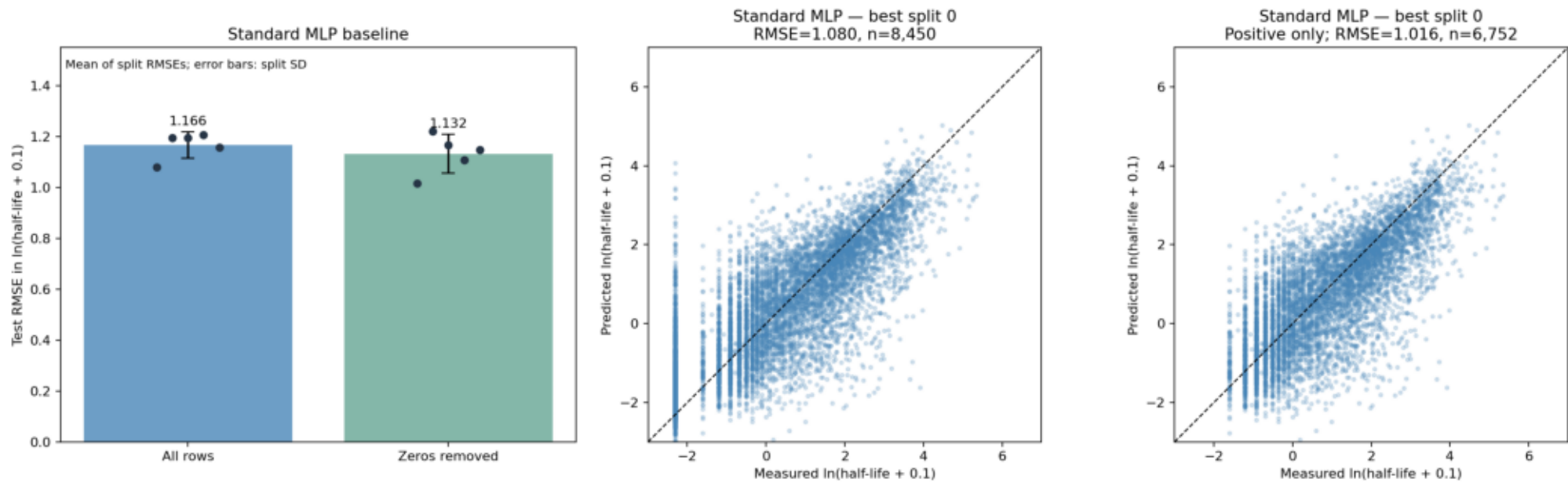
Target: $\ln(\text{half-life} + 0.1)$. Same saved split i for each model. Lower RMSE is better.

Best test scatter plots: measured zeros removed



Target: $\ln(\text{half-life} + 0.1)$. Same saved split i for each model. Lower RMSE is better.

Standard MLP baseline: all rows versus zeros removed



Target: $\ln(\text{half-life} + 0.1)$. Same saved split i for each model. Lower RMSE is better.

Data sources and reproducibility

Repository: [milorepossi/peptide-HLA-stability](#)

Standard MLP: RESULTS/metrics.json and mlp_split_i.pt

Fine-tuned model: RESULTS/ranking_finetuned/ranking_finetuned/

Frozen initialization: RESULTS/ranking_finetuned/A3_L33_flatten_localrun/

Test data: DATA/testing_i.npz, $i = 0, 1, 2, 3, 4$

Standard MLP test predictions were regenerated from each matching checkpoint and checked against reported MSE. Fine-tuning predictions were matched by source_row and measured target values.

The fine-tuning project's saved splits differ slightly from testing_data splits. This report consistently uses the fine-tuning project's test sets.

Positive-only comparison: remove rows with measured half-life = 0; retain the original models and original log target transformation.

One run per model and split; no statistical significance claim is made.